Yawar Badri

Systems Engineering | Hardware Architecture | Power Electronics

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Professional Summary

Electrical engineer and systems architect with experience in power electronics, embedded systems, hardware design, and engineering tools for space and terrestrial applications. Work spans system architecture, PCB design, firmware, verification, and commissioning, including satellite power systems, grid-connected converters, energy storage, and test equipment.

Core Competencies

- **Design & analysis:** KiCad, Altium, Cadence, PCB layout, MATLAB/Simulink, LTspice, Qspice, FMECA, WCCA
- **Firmware & software:** C, C++, C#, Python, JavaScript, STM32, ARM Cortex, embedded DSP
- **Verification:** EGSE, hardware-in-the-loop (HIL), oscilloscopes, logic analyzers, LCR, DMM, power quality analysis

Experience

Pixxel Space **2022 – Present**
Senior Electrical Engineer *Space / High-Rel Electronics*

– Architected end-to-end electrical power systems (EPS) for LEO satellites, from sizing and component selection through assembly. Designed buck power tiles for up to 100 V, achieving approximately 800 W/in^3 power density.

– Led small teams through design, test, and commissioning of Firefly satellites, coordinating with thermal, mechanical, AOCS, and AIT teams.

– Built local component search tools and databases for COTS part selection; contributed to a wider sourcing effort that reduced selected lead times from over six months to one to four weeks.

– Developed electrical ground support equipment (EGSE), including a modular 500 W solar string simulator and multichannel power loggers.

Kashmir University **2021 – 2025**
Technical Advisor & Mentor *Hardware Startups & Prototyping*

– Advised students and early-stage hardware teams on system architecture, prototyping, and embedded integration.

– Mentored PhD students in hardware design and conducted embedded systems workshops.

NIT Srinagar **2022**
Project Associate & Mentor *Power Systems*

– Led hardware architecture and systems integration for an experimental 10 kW grid-tied inverter testbed.

– Mentored students in hardware and power electronics.

University of Kashmir (Sponsored by J&K Govt.) **2021**
Research Analyst / Associate *EV Infrastructure Analysis*

– Analyzed distribution constraints for EV infrastructure and translated technical findings into state policy recommendations.

Selected Hardware Projects

- **10-channel power logger:** Designed a bidirectional DC logger rated up to 100 V, including analog front end, STM32 USB interface, firmware, calibration, and desktop plotting application.
- **DACard-Basic HIL interface:** Built hardware, embedded firmware, host software, and Python libraries for a hardware-in-the-loop test platform.
- **Three-phase grid-tied inverter:** Designed the board layout for power conversion supporting active and reactive power control research.
- **Ground energy storage BMS:** Designed and deployed battery management systems for ground-based energy storage.

Software & Engineering Tools

- **WayriCAD (formerly KiWay):** Open-source KiCad plugin suite for PCB design and analysis, including fanout routing, via stitching, BOM editing, trace RLC, and power and signal integrity checks.
- **CATS:** CAN-based automated hardware testing with a command, read, verify, and record workflow through PCAN hardware.
- **sigLib:** C++ DSP library for real-time waveform conditioning, smoothing, and analysis on embedded targets.
- **Custom engineering applications:** Developed transformer design, data acquisition, load-profile, and billing applications.

Interactive Engineering Utilities

- Built browser-based calculators for SMPS topologies, loop compensation, motor dynamics, PV yield, battery configuration, filters, trace sizing, and magnetics.
- Developed a Web Serial interface for live hardware data monitoring and plotting.

Education

Lovely Professional University (LPU) **2022**
Master of Technology (M.Tech) in Electrical Engineering (Power & Renewable Energy) *Grade: A*

University of Kashmir **2019**
Bachelor of Technology (B.Tech) in Electrical Engineering *Grade: A*

Selected Publications

- *Ultracapacitor as selectable energy buffer in electric vehicle application* (ScienceDirect/Elsevier, 2023)
- *Experimental Implementation of Optimally Tuned Multifunctional RLMLS Driven VSc Tied to Single Phase Voltage Weak Distribution Grid* (Springer, 2023)
- *Application of V2V Energy Sharing in Electric Vehicles with Source Switching* (IEEE Xplore, 2022)
- *A New Modular Multilevel Converter Topology Using Flying Ultra-capacitor and Cascaded H-bridges* (Springer, 2021)